

SJF vs Priority Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	SJF vs Priority Comparison Project	Team Number	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
1					
2					
3					
4					
5					
6					
7					

Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	SJF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	Priority Scheduling	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.

Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	SJF selection is based on the shortest available burst time.	
☐	Priority rule is stated clearly and applied consistently.	
☐	At least one workload shows conflict between short burst time and high priority.	
☐	Comparison tables use the same dataset for both	

	algorithms.	
□	Conclusion explains efficiency versus urgency trade-off.	

Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Assume **preemptive** SJF and Priority Scheduling.
- Discuss efficiency versus urgency when short jobs and high-priority jobs conflict.
- Explain whether either policy causes unfair delay.